

Lewis structure &

(2 < 8 / 8 gives 15)

bonding
 ionic covalent

H2O_GROUP

Na :- $1s^2 2s^2 2p^6 3s^1$ 1 electron

Cl :- $1s^2 2s^2 2p^6 3s^2 3p^5$ 7 electrons

stable

NaCl ionic bond.

N :- $1s^2 2s^2 2p^3$ 5 electrons

both of them needs electrons

O :- $1s^2 2s^2 2p^4$ 6 electrons

No covalent bond
 "strong"

Ionic $\Rightarrow$ metal + nonmetal

covalent $\Rightarrow$ none + none "near each other"

Lattice Energy



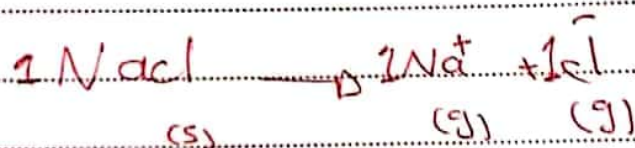
► Subject : Lattice energy

Amount of energy required to completely separate 1 mole of solid ionic compound into gaseous state

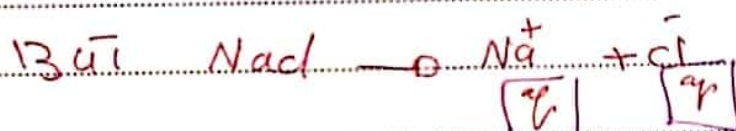
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There's 6 conditions or

1. 1 mole
2. Solid
3. Ionic compound
4. gaseous
5. separate
6. Ions



all of them are here!



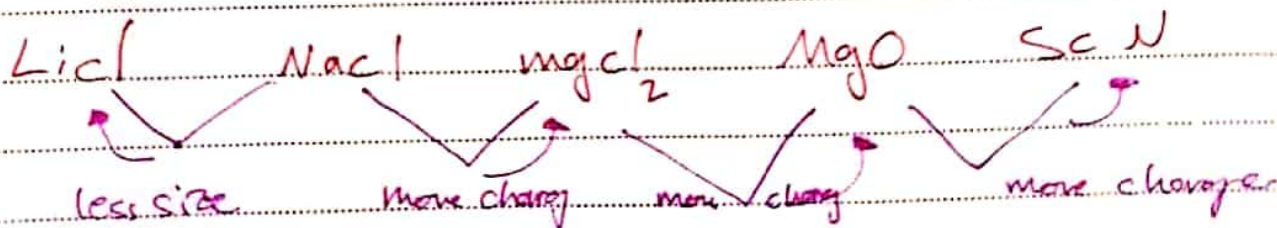
not lattice $\times \mathbb{P}$ only ΔH

reactions $\left\{ \begin{array}{l} \rightarrow \text{separation} \\ \rightarrow \text{compensation} \end{array} \right.$



$$\Delta H = 2 \Delta H_{\text{lattice}}$$

* lattice E increases as charge increases and size decreases



1, 2, 3, 4 electrons	large size	small	$\begin{matrix} +2 \\ \text{mg} \\ 12 \\ + \\ \text{Al} \end{matrix}$
charge			

Subject :

→ one factor may be more effective than the others.

→ charge is more effective than size here

* Which of the following compounds has the highest melting point?

1. BeH_2

melting point: ZnCl_2 is 270°C ; AlCl_3 is 190°C ; BH_3 is a gas. ZnCl_2 is a solid.

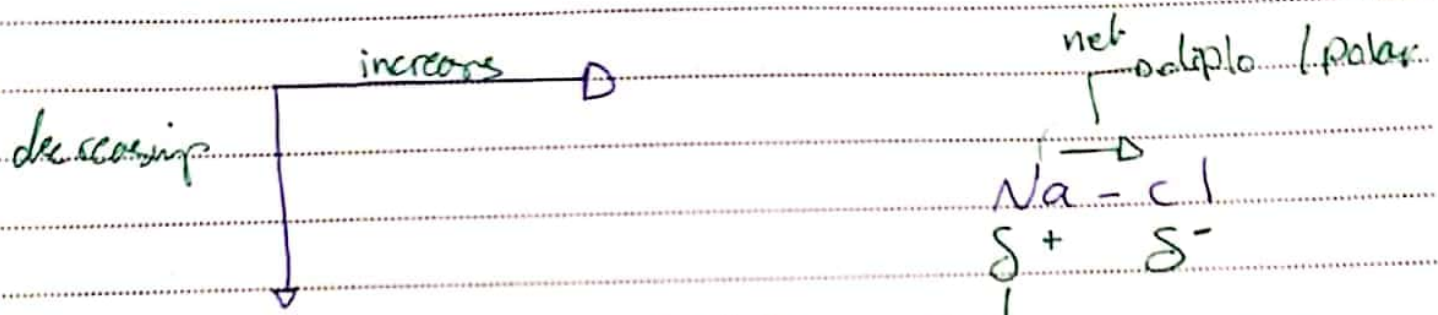
melting point proportional with $\Delta H_{\text{lattice}}$.

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electronegativity

Ability of [an atom in a molecule] to attract electrons to itself

The difference



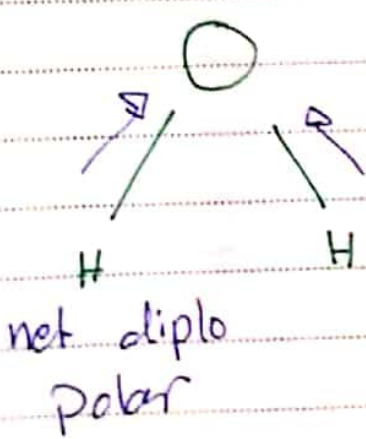
H2O_GROUP

partially not full

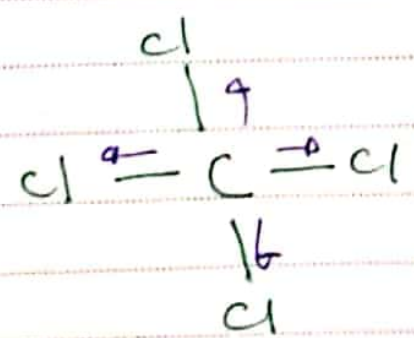


dipole moment
 $\downarrow$
Polar

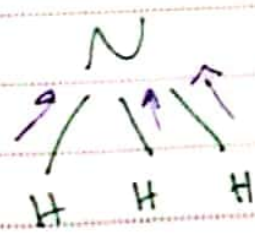
الغرض بلفظي يعني
net dipole moment
 $\downarrow$
not Polar



not net dipole
so it's not
polar



net dipole
Polar



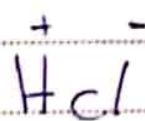
$$\text{Dipole moment} = Q \times r$$

charge distance between atoms

$$1 \text{ D} = 3.34 \times 10^{-30} \text{ C} \cdot \text{m}$$

مثال ١

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$$+1 \propto 1.6 \times 10^{-19} \text{ coulomb}$$

charge of an atom is unbalanced

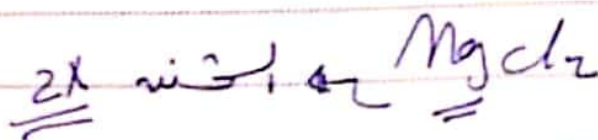
* Dipole moment for HCl is 1.08 D , if the distance H and Cl is $1.27 \text{ (Å)} \times 10^{-10}$

→ What's the magnitude of charge on both H and Cl.

• $Q = \frac{\mu}{r}$...

$$\mu = Q \times r \quad \frac{1.08 \text{ D} \times 3.34 \times 10^{-30}}{1 \text{ D}} = Q \times 1.27 \times 10^{-10} \text{ m}$$

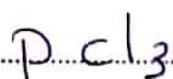
$$Q = \frac{\boxed{} \text{ coulomb}}{\text{electron charge closer}} = \boxed{} \text{ unbalanced}$$



► Subject: Drawing Lewis structure

Tips:-

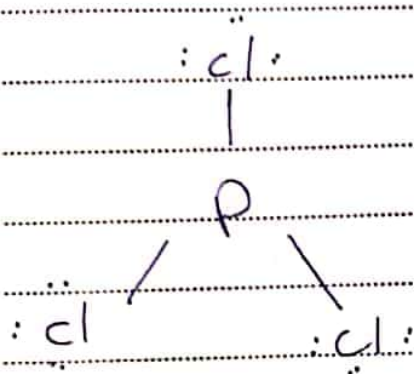
- ① Lewis electrons
- ② real electrons
- ③ subtraction
- ④ bonds



Lewis :- $4 \times 8 = 32 e^-$ ()

Real :- $5 + \frac{21}{2} = 26 e^-$

$6 e^- \Rightarrow 3 \text{ bonds}$



Lewis: $2 + 16 = 18$

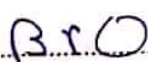
Real: $\frac{10}{2}$

$8 e^- \Rightarrow 4 \text{ bonds}$

H2O_GROUP

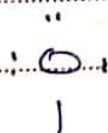
The only choice as (H) gets only 2 electrons $\leftarrow \text{H} - \text{C} \equiv \text{N} :$ (4 bonds, 1 lone electron)

Pounding
electron



Lewis :- $3 \times 32 e^-$ gain electron

Real :- $7 + (6 \times 3) + 1 = 26$

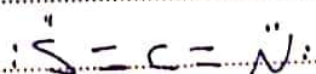


$6 e^- \Rightarrow 3 \text{ bonds}$

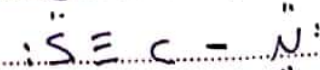


Lewis: 24

Real: $6 + 4 + 5 + 1 = 16$



$8 e^- \Rightarrow 4 \text{ bonds}$

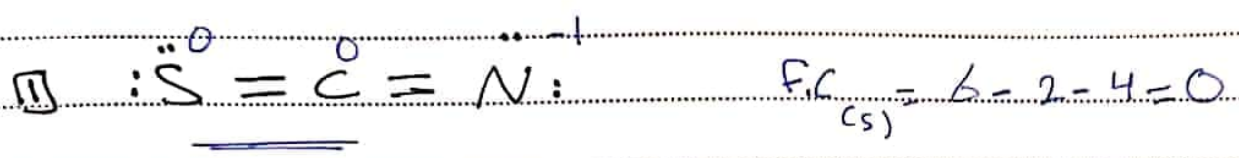


one of them is

more stable

Subject : formal charge

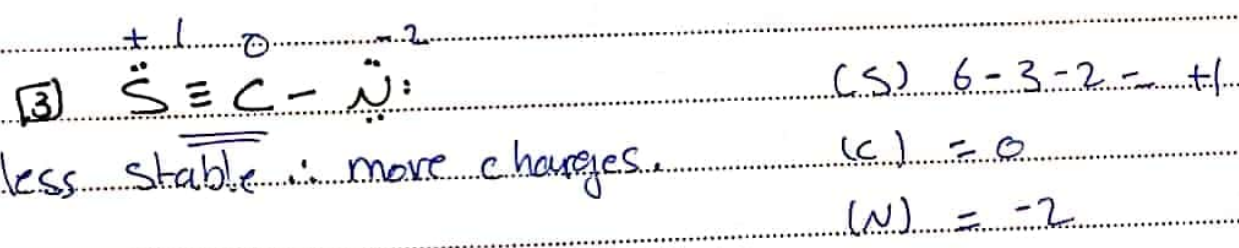
Formal charge = $\text{Valence e's} - \frac{1}{2} \text{bonding e's} - \text{non bonding e's}$



The most stable is less charges and (N) is more electronegativity than the (S). $\text{F.C.}_{(\text{C})} = 4 - 4 - 0 = 0$
 $\text{F.C.}_{(\text{N})} = -1$

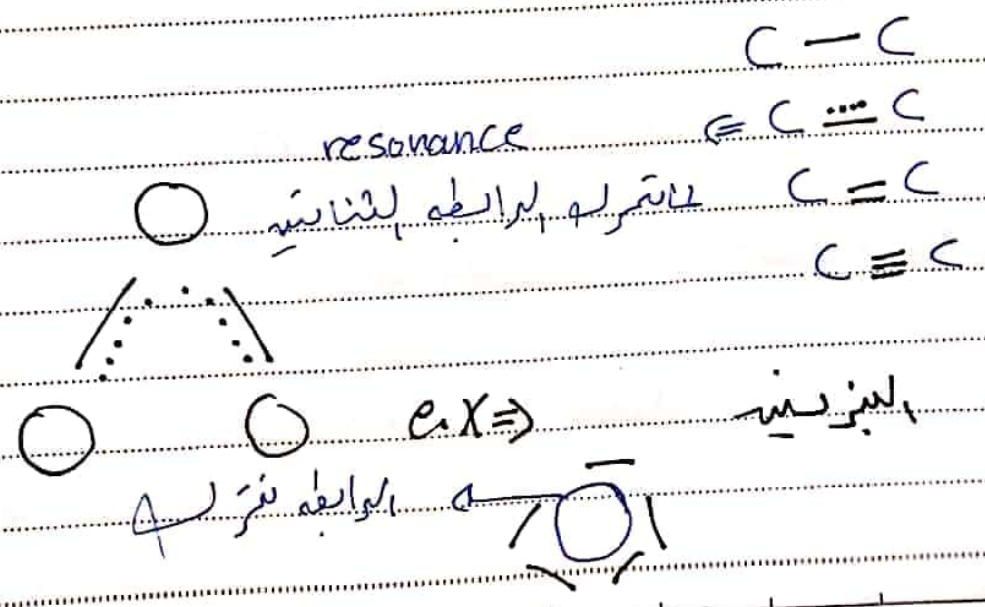


H2O_GROUP

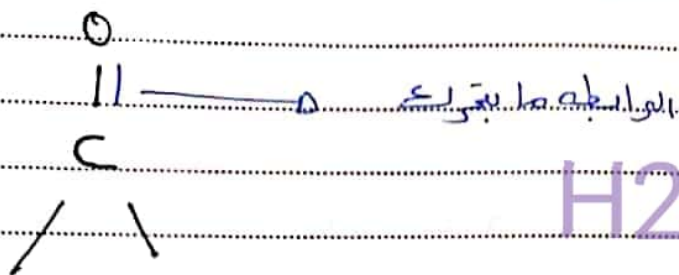


Q: Which one is the most stable? The first one why?
 الذرة ما بعيب لحنه الجسيم + أكثر كبره

* O_3
 Lewis: 24
 real: $3 \times 6 = 18$
 $6 \Rightarrow 3 \text{ bonds}$



Subject: Lewis exceptions.



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* bond length $\propto$ 1 bond strength

هذا يعني أن قوة الرابطة تتناسب عكسياً مع طولها (The longer the bond, the weaker it is).

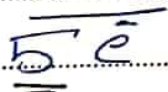
double / single $\rightarrow$ resonance $\leq$ في الحقيقة هي 6 أشكال من الرنين (In reality, there are 6 resonance structures).

Expection

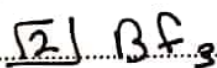


Lewis: 16

real: 11



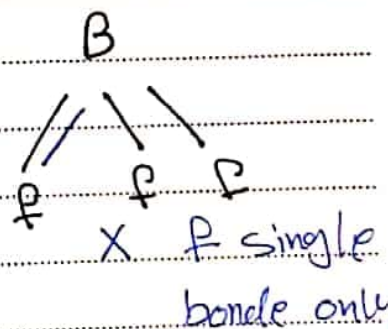
Odd number



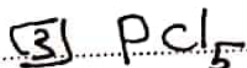
Lewis: 32

real: 24

$8 \Rightarrow$ 4 bonds



[1] odd number of electron (NO)



Lewis: 48

real: 40

$8 \Rightarrow$ 4 bonds

BUT we have 5 Atoms

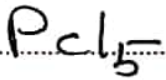
بما أن 4 $\rightarrow$ 5 ذرات (Since 4 $\rightarrow$ 5 atoms)

[2] some atoms make single bond only (BF₃)

[3] number of bonds is less than number of atoms. (PCl₅)

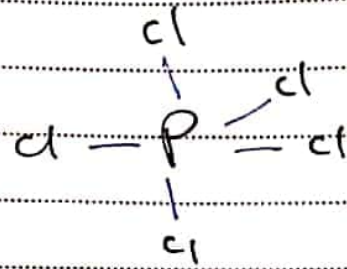
Subject: Exception molecule

طريقة معالجة موانع الفشل لدى لويس



$5 + 5 = 10e^- \Rightarrow 5 \text{ bonds}$
more than octet

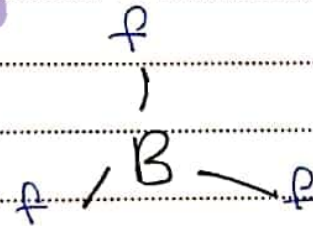
Valence عدد إلكترونات الترابط
مع الإلكترونات الحرة



H2O_GROUP

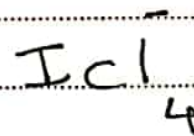


$3 + 3 = 6 \Rightarrow 3 \text{ bonds}$
less than octet



* ~~an~~ octet rule :- (8) if # e's is more or less than (8)

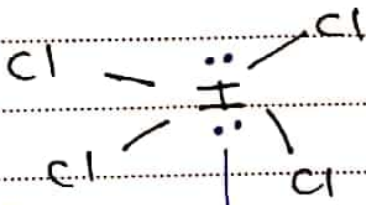
$7 + 4 + 1 = 12e^- \Rightarrow 6 \text{ bonds}$
method 1



Lewis: 40

real: 36

$4e^- \Rightarrow 2 \text{ bonds} \times$



Double bond

NOTES

1. method 1

2. method 2 Double / Triple

(Only Lewis)

الزوجين غير الترابطية
lone pairs
non bonding pairs